Task 8 : AWS RDS Setup and Grafana Cloud Dashboards for Data Visualization

Task: AWS RDS Setup with Data Visualization in Grafana Cloud

Objective

Set up an AWS RDS database using the provided dataset and create monitoring and analytics dashboards in Grafana Cloud.

Note : **Please create a Grafana Cloud account** (free tier is fine) and configure the necessary integrations.

Grafana Cloud

Dataset

Use the following database file:

datasets/world.db at main · gramener/datasets

This database contains sample world data (countries, cities, etc.) and will be used to populate RDS with structured dummy data.

Task Requirements

1️⃣ Create AWS RDS Instance

- Create an **Amazon RDS instance** (MySQL or PostgreSQL preferred).
- Configure:
  - Appropriate instance type
  - Networking and security groups
  - Enable **Performance Insights**
- Create a database inside RDS.
- Import the data from `world.db` into the RDS instance.
- Validate tables and data using SQL queries.

2️⃣ Create Grafana Cloud Account

- Sign up for a **Grafana Cloud** account
- Configure:

- CloudWatch integration for infrastructure metrics
- Direct database connection (MySQL/PostgreSQL datasource)

3️⃣ Dashboard 1 – RDS Infrastructure Metrics

Create a dashboard to monitor **RDS infrastructure performance**, including:

- CPU utilization
- Memory usage
- Storage usage
- Database connections
- Read/Write IOPS
- Performance Insights metrics (DB load, top queries)

(Use CloudWatch and Performance Insights integration.)

4️⃣ Dashboard 2 – Database Insights

Create a dashboard showing insights from the imported dataset, such as:

- Total number of cities
- Country-wise city count
- Population-based metrics (if available)
- Top 10 countries by number of cities

Use SQL queries in Grafana to fetch and visualize this data.

⭐ Bonus Task (Automation + Query Load Simulation)

Create a small automation setup to simulate database activity:

- Write a script that runs **random SQL queries** against the RDS database.
- Schedule the script using a **cron job** (e.g., every 5 minutes).
- Queries can include:
  - Random city lookups
  - Country-wise aggregations
  - Population-based filters
- Ensure queries generate measurable load.

- Visualize the impact in the **Performance Insights metrics dashboard** (e.g., DB load, query activity trends).

Expected Deliverables

Submit the following:

1. Requirement analysis (Confluence link)
2. Technical documentation (Confluence link)
3. Prepare a short document explaining difference between setting up infrastructure dashboard ,analytics dashboard and performance insights dashboard from a DB (Confluence link)
    - Also include a short video (2–3 mins) explaining the same documentation.

Time to Complete

5–6 Hours